Junyi (Conny) Zhou

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Education

- **Harvard University** — Boston, MA
  M.S. in Health Data Science — *August 2025 – March 2027*
  - Program requirements complete December 2026 (commencement March 2027); available for full-time start January 2027
  - Coursework: Deep Learning, AI in Medicine, Healthcare Machine Learning, Data Science
- **Emory University, College of Arts and Sciences** — Atlanta, GA
  B.S. in Applied Mathematics and Statistics; Minor in Computer Informatics; GPA: 3.925 — *August 2021 – May 2025*
  - Coursework: Probability and Statistics, Machine Learning, NLP, Data Structures, Database Systems

Technical Skills

- **Languages**: Python, C++, Java, SQL, R, JavaScript
- **ML & Deep Learning**: PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, LightGBM, Hugging Face Transformers
- **LLM & Applied AI**: RAG, agentic pipelines, prompt engineering, embeddings & vector search, OpenAI API, MongoDB Atlas Vector Search
- **Data & Compute**: PostgreSQL, MySQL, MongoDB, Spark, Hadoop, SLURM/HPC, multi-GPU job scheduling
- **Cloud**: AWS (SageMaker, Lambda, EC2, S3, Redshift, Elastic Beanstalk, CloudFront), Snowflake, BigQuery
- **Backend & DevOps**: Flask, REST APIs, React, Docker, Auth0, Git
- **Certifications**: AWS Certified Cloud Practitioner, AWS Certified AI Practitioner, AWS Machine Learning Associate

Clinical AI Experience

- **Airway Management Simulation Chatbot** [Live Demo] — Atlanta, GA
  Full-Stack Engineer, Scrum Master — Emory Pediatric Hospital — *January 2024 – August 2025*
  - Architected a clinical **agentic LLM pipeline** pairing **RAG** retrieval over **MongoDB Atlas Vector Search** with **prompt engineering** and context-window management to generate multi-turn training scenarios from **OpenAI** models.
  - Built a **HIPAA-conscious** full-stack platform (**Python/Flask** REST API, React, MongoDB, **Auth0**) deployed on **AWS** Elastic Beanstalk and CloudFront, tracking resident performance metrics.
  - Led an Agile team of 6, running code reviews and folding clinician feedback into successive releases.
- **Cross-Species Drug Translation using Optimal Transport & Generative AI** — Boston, MA
  Graduate Researcher — Mooney Lab & Alvarez-Melis Lab, Wyss Institute — *February 2026 – Present*
  - Built **Optimal Transport** models (**VAEs**, **unbalanced OT**) predicting animal-to-human trial translation from **scRNA-seq** atlases, with **scanpy** pipelines automating cell-type labeling and dataset integration.
  - Ported a 2018 CPU-only **TensorFlow** reference implementation to **PyTorch**, reimplementing the **Input Convex Neural Networks** behind the optimal-transport dual and enabling GPU training on Harvard's **FASRC Cannon** cluster.
  - Benchmarked **CellOT** against **scGen** on cross-species perturbation prediction, measuring 0.85–0.90 R^2 with the source species in training versus 0.65–0.67 on the held-out-species split.
  - Established through **paired single-factor ablations** that gene-space choice reverses with split — 6,619 orthologs wins in-distribution while 1,000 HVGs wins on every out-of-distribution metric — ruling out a split-independent default.

Machine Learning Projects

- **Pneumonia Detection from Chest Radiographs** [GitHub] — Boston, MA
  Harvard BST-261 (Kaggle competition) — *Spring 2026*
  - Diagnosed a 14-point generalization gap on a **Kaggle** chest-radiograph benchmark — 99% validation against 85% leaderboard — and traced it to memorization of a 4,700-image training set.
  - Closed the gap to 96% with domain-aware augmentation (**Mixup** $\alpha = 0.2$, RandomResizedCrop, RandomErasing, affine jitter), excluding vertical flips that would render chest radiographs anatomically implausible.
  - Found that freezing **DenseNet-121**'s early layers outweighed architecture choice, beating ResNet-50, ConvNeXt-Tiny, EfficientNet-B0, and a three-model ensemble to reach a 0.959 public score.
- **AlphaFold Protein–Nucleic Acid Interaction Prediction** [GitHub] — Atlanta, GA
  Data Analyst — Bou-Nader Lab, Emory Dept. of Biochemistry — *February 2025 – August 2025*
  - Engineered a pipeline integrating **AlphaFold-Multimer** and **AlphaFold3** to predict protein–nucleic acid interactions from **MSA** data.
  - Automated high-throughput inference across GPU nodes with **SLURM** parallel scheduling, tuning batch size to raise GPU utilization on shared **HPC** clusters.
  - Built **Python** tooling (PyMOL, ChimeraX) ranking structures by **pLDDT** and **PAE** score, shipped as a documented CLI so non-technical researchers could run inference and triage results independently.